Customer Churn Prediction Report

Executive Summary

This report presents a machine learning analysis designed to predict telecom customer churn. The objective is to identify customers most likely to leave the service and support targeted retention strategies. The final model ranks customers by churn risk and identifies a high-risk segment with significantly higher churn probability, enabling more efficient retention actions.

Dataset Overview

The dataset contains 7,043 customers and 21 original variables including demographics, services, billing details, and contract type. The churn rate is approximately 27%, meaning the dataset is moderately imbalanced.

Metric	Value
Total customers	7,043
Original features	21
Features after preprocessing	23
Churned customers	1,869
Non-churned customers	5,174

Exploratory Data Analysis

EDA revealed strong churn patterns across contract type, payment method, tenure and pricing.
Month-to-month contracts and electronic check payments show the highest churn rates.
Customers with shorter tenure and higher monthly charges also exhibit higher churn risk.

Modeling Approach

The dataset was split into training and validation sets using a stratified 75/25 split (5,282 training /1,761 validation). A baseline dummy classifier was first evaluated, followed by Logistic Regression and Random Forest models. Performance was measured using Average Precision (PR-AUC) and ROC-AUC to evaluate ranking quality on an imbalanced dataset.

Model Performance

Model	Average Precision	ROC-AUC
Baseline (Dummy)	0.274	0.520
Logistic Regression	0.634	0.846
Random Forest	0.641	0.843

Both ML models significantly outperform the baseline. Random Forest achieved the highest Average Precision and was selected as the final model.

Final Model Results

Metric	Value
Accuracy	0.77
Precision (Churn)	0.56
Recall (Churn)	0.69
F1 Score (Churn)	0.62

The model correctly identifies roughly 69% of customers who churn. In retention use cases recall is particularly important because missing a churner is more costly than contacting a customer who would have stayed.

Key Drivers of Churn

Feature	Importance
Contract	0.173
Tenure	0.164
TotalCharges	0.134
MonthlyCharges	0.125
InternetService_Fiber optic	0.075
PaymentMethod_Electronic check	0.053
InternetService_No	0.032
PaperlessBilling	0.030

Contract type and tenure are the strongest predictors of churn. Customers on month-to-month contracts and customers early in their lifecycle are significantly more likely to leave. Pricing and service bundle factors also influence churn.

Risk Segmentation

Risk Tier	Customers
High	438
Medium	400
Low	923

The top 20% highest-risk customers have a churn rate of 67%, more than double the overall churn rate (~27%). This shows the model successfully concentrates risk into a manageable group for targeted retention campaigns.

Business Recommendations

• Encourage longer-term contracts with incentives.
• Focus retention efforts on new customers with low tenure.
• Review pricing and value perception for high monthly charge customers.
• Promote service bundles such as Tech Support and Online Security.
• Encourage automatic payment methods instead of electronic checks.

Conclusion

The churn prediction model provides a practical tool for proactive customer retention. By ranking customers by risk and identifying key churn drivers, the company can prioritize interventions and reduce revenue loss from churn.